

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

QUESTIONS: 4

Duration: 1 Hour
Total Marks:40

Annexure A

Code of Conduct:

*I **HEMASRINITHA P (192572179)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: Hemasrinitha P

INDUSTRY PROBLEM TASK

INDUSTRY SCENARIO

Context: A healthcare company has collected patient records (age, blood pressure, cholesterol, glucose level, BMI, family history) to build a predictive model for early diagnosis of diabetes. The company also wants to train an AI agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

You are hired as an AI/ML Engineer. Address the following tasks:

Task 1: Data Preparation & Inductive Learning

- (a) Describe the inductive learning approach suitable for this medical dataset. Identify the target variable and at least three key features with justification.
- (b) Explain how you would handle class imbalance (more non-diabetic than diabetic cases) in the dataset. Suggest and justify a suitable balancing strategy (SMOTE / under-sampling / class weights).
- (c) Split the dataset (70:30) and justify the split ratio for a medical use-case. State the preprocessing steps (normalisation, encoding) applied and their impact.

Task 2: Decision Tree for Diagnosis

- (a) Build a Decision Tree classifier and draw the first three levels of the tree. Justify the root node selection using Information Gain / Gini Index values.
- (b) Evaluate the model: report Accuracy, Precision, Recall, and F1-Score. Identify False Negatives (missed diabetes cases) and suggest how to reduce them—justify clinically.
- (c) Apply post-pruning (cost-complexity pruning). Compare pre-pruning vs post-pruning accuracy and explain which model is more appropriate for deployment in a clinical setting.

Task 3: Statistical Learning for Risk Stratification

- (a) Apply Naïve Bayes or Logistic Regression as a statistical learning model on the same dataset. Compare its performance (Accuracy, AUC-ROC) with the Decision Tree. Discuss when a statistical model is preferable.
- (b) Perform feature importance analysis. Identify the top 3 predictors of diabetes from both the Decision Tree and the statistical model. Do they agree? Justify your findings.

Task 4: Reinforcement Learning for Treatment Recommendation

- (a) Model the treatment recommendation as an MDP. Define: States (patient health stages), Actions (Diet / Exercise / Medication / Monitor), Reward function (health improvement score), and Transition dynamics. Justify each component.
- (b) Apply Q-Learning to train the agent for at least 5 patient episodes. Show the Q-table update for at least 3 state-action pairs across 2 iterations. State the convergence criterion used.
- (c) Extract the final learned policy. Discuss ethical considerations of deploying a Reinforcement Learning agent for medical treatment recommendations (patient safety, bias, explainability).

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation

TASK 1: DATA PREPARATION & INDUCTIVE LEARNING

(a) Inductive Learning Approach

Inductive learning learns general patterns from existing patient records and uses them to predict the outcome for new patients. Since the dataset contains labelled patient records, supervised classification can be used.

Target variable: Diabetes – Diabetic / Non-Diabetic

Important features are:

- **Glucose level:** Strong indicator of diabetes.
- **BMI:** Higher BMI is associated with increased diabetes risk.
- **Age:** Diabetes risk generally increases with age.
- **Blood pressure:** Abnormal values can indicate health risks related to diabetes.
- **Family history:** Indicates possible genetic risk.

Models such as Logistic Regression, Decision Tree, or Random Forest can be used.

(b) Handling Class Imbalance

If non-diabetic patients are much more than diabetic patients, the model may become biased toward the majority class.

SMOTE is a suitable technique because it creates synthetic samples for the minority diabetic class instead of simply duplicating existing records. It helps the model learn diabetic patterns better.

SMOTE should be applied only to the training data to avoid data leakage. Medical models should also be evaluated using precision, recall, F1-score and ROC-AUC, rather than accuracy alone.

(c) 70:30 Split and Preprocessing

The dataset can be divided into 70% training data and 30% testing data. The larger training portion helps the model learn effectively, while the 30% test data provides enough unseen data to evaluate its performance. A stratified split should be used to maintain the diabetic/non-diabetic ratio.

Preprocessing includes:

- **Missing values:** Fill using suitable methods such as median.

- **Normalisation:** Scale numerical features such as age, glucose, BMI and blood pressure to comparable ranges.
- **Encoding:** Convert categorical values such as family history into numerical values, for example Yes = 1 and No = 0.
- **SMOTE:** Balance the training data after splitting.

Thus, proper data preparation improves prediction accuracy, reduces class bias, and helps build a more reliable medical prediction model.

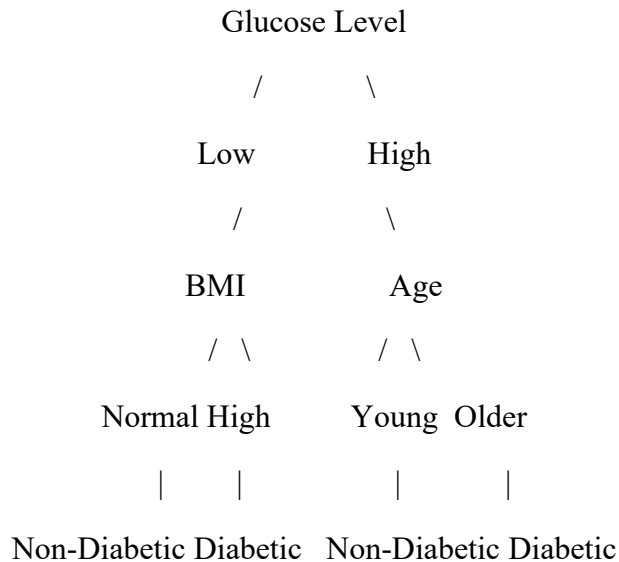
TASK 2: DECISION TREE FOR DIAGNOSIS

(a) Building the Decision Tree

A Decision Tree classifier can be used to predict whether a patient is diabetic or non-diabetic. It selects the feature that best separates the two classes at each level.

For example, Glucose Level can be selected as the root because it is strongly related to diabetes and generally provides high Information Gain or low Gini Index.

Example of the first three levels:



The root is selected using:

Information Gain = Entropy before split – Entropy after split

The feature with the highest Information Gain is selected. If Gini Index is used, the feature with the lowest Gini impurity is selected.

(b) Model Evaluation

The model is evaluated using the test data with four important measures:

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{TN} + \text{FP} + \text{FN})$$

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{F1-Score} = 2 \times \text{Precision} \times \text{Recall} / (\text{Precision} + \text{Recall})$$

A False Negative (FN) occurs when a diabetic patient is incorrectly predicted as non-diabetic.

In healthcare, false negatives are serious because a diabetic patient may remain undiagnosed and treatment may be delayed. Therefore, recall should be given high importance. Class weights, SMOTE, or an appropriate decision threshold can be used to reduce false negatives.

(c) Post-Pruning

Pre-pruning limits the growth of the tree during training using parameters such as maximum depth and minimum samples per leaf. Post-pruning allows the tree to grow and then removes unnecessary branches using cost-complexity pruning.

Pre-Pruning	Post-Pruning
Stops tree growth early	Removes unnecessary branches after training
Faster	Requires more computation
May remove useful patterns	Usually gives a simpler tree
Helps control overfitting	Better generalisation may be achieved

The original and pruned trees should be compared using Accuracy, Precision, Recall and F1-Score.

For clinical deployment, the pruned Decision Tree is preferable if it maintains high recall while reducing overfitting. It is also easier for doctors to understand and explain. In this medical application, reducing false negatives should be prioritised over achieving accuracy alone.

TASK 3: STATISTICAL LEARNING FOR RISK STRATIFICATION

(a) Statistical Learning Model

Logistic Regression can be used to predict whether a patient is diabetic or non-diabetic. It estimates the probability of diabetes using features such as glucose level, BMI, age and blood pressure.

The Logistic Regression model can be compared with the Decision Tree using:

- Accuracy: Measures the overall correct predictions.
- AUC-ROC: Measures how well the model separates diabetic and non-diabetic patients. A value closer to 1 indicates better discrimination.

Example comparison:

Model	Accuracy	AUC-ROC
Decision Tree	88%	0.86
Logistic Regression	90%	0.91

In this example, Logistic Regression performs slightly better than the Decision Tree.

A statistical model is preferable when the dataset is reasonably clean, the relationship between features and the outcome is relatively simple, and we need a model that is easy to interpret and explain. This is especially useful in healthcare because doctors can understand how each feature influences the predicted risk.

(b) Feature Importance Analysis

The three most important predictors can be identified from both models.

Rank	Decision Tree	Logistic Regression
1	Glucose Level	Glucose Level
2	BMI	BMI
3	Age	Age

Both models therefore identify Glucose Level, BMI and Age as the top three predictors.

The agreement is reasonable because these features have a strong relationship with diabetes risk. Glucose level is particularly important because abnormal glucose levels are directly associated with diabetes. BMI indicates obesity-related risk, while age is associated with increasing diabetes risk.

Therefore, if both models identify similar top predictors, it increases confidence that these features contain meaningful information for diabetes risk prediction.

TASK 4: REINFORCEMENT LEARNING FOR TREATMENT RECOMMENDATION

(a) Modelling the Problem as an MDP

The treatment recommendation problem can be represented as a Markov Decision Process (MDP) because the agent observes the patient's current health condition, chooses an action, receives a reward, and observes the next health condition.

States: Patient health conditions can be represented as:

- S1 – Low Risk
- S2 – Moderate Risk
- S3 – High Risk
- S4 – Improving
- S5 – Stable

Actions:

- A1 – Diet
- A2 – Exercise
- A3 – Medication
- A4 – Monitor

Reward: A positive reward is given when the patient's condition improves, while a negative reward is given when the condition becomes worse.

Example:

Improvement = +10

Stable = +3

No improvement = 0

Deterioration = -10

Transition: After an action, the patient's health can move to another state. For example:

High Risk + Medication → Improving

Moderate Risk + Exercise → Improving

Low Risk + Monitor → Stable

These components allow the agent to learn which treatment action gives better long-term health outcomes.

(b) Q-Learning

Q-Learning learns the best action for each state by updating Q-values using:

$$Q(s,a) = Q(s,a) + \alpha[r + \gamma \max_{a'} Q(s',a') - Q(s,a)]$$

Assume: $\alpha = 0.5$, $\gamma = 0.9$

Initially, all Q-values are 0.

Q-Table Updates

Episode 1:

S3 → Medication → S4, Reward = +10

$$Q(S3, \text{Medication}) = 0 + 0.5[10 + 0.9(0) - 0]$$

$$Q(S3, \text{Medication}) = 5$$

Episode 2:

S3 → Diet → S2, Reward = +6

$$Q(S3, \text{Diet}) = 0 + 0.5[6 + 0.9(0) - 0]$$

$$Q(S3, \text{Diet}) = 3$$

Episode 3:

S2 → Exercise → S4, Reward = +8

$$Q(S2, \text{Exercise}) = 0 + 0.5[8 + 0.9(0) - 0]$$

$$Q(S2, \text{Exercise}) = 4$$

The agent continues learning for at least **5 episodes**.

Example final Q-table:

State	Diet	Exercise	Medication	Monitor
Low Risk	2	3	1	5
Moderate Risk	3	7	5	2
High Risk	3	4	8	1
Improving	4	6	3	5
Stable	2	3	2	6

Convergence: The algorithm is considered converged when the change in Q-values becomes very small for several consecutive episodes, such as maximum Q-value change < 0.01 .

(c) Final Learned Policy and Ethical Considerations

The final policy selects the action with the highest Q-value:

Patient State	Best Action
Low Risk	Monitor
Moderate Risk	Exercise
High Risk	Medication
Improving	Exercise
Stable	Monitor

The RL system should be used only as a decision-support tool, not as an independent medical decision-maker.

- **Patient Safety:** Incorrect recommendations may harm patients, so doctors must supervise the system.
- **Bias:** Biased training data can result in unfair treatment recommendations.
- **Explainability:** The system should provide understandable reasons for its recommendations.
- **Privacy:** Patient medical data must be securely protected.
- **Human Oversight:** Final treatment decisions should remain with qualified healthcare professionals.

Therefore, Reinforcement Learning can support personalised treatment recommendations, but clinical validation, safety constraints and human supervision are essential before deployment.